

QUESTIONS

Q1

1. Find the bugs, include the missing lines in the following snippet.
1 mark

Q2

2. Buggy Code in an array
1 mark

Q3

Identify an errors in the following snippet.
1 mark

3/3 answered

Q1 1. Find the bugs, include the missing lines in the following snippet.

1 mark

```
int linearSearch(int arr[] int n, int key) {
    for(int i = 0; i <= n; i++) {
        if(arr[i] == key)
            return i;
    }
    return -1;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int linearSearch(int arr[], int n, int key) {
3     for (int i = 0; i < n; i++) {
4         if (arr[i] == key) {
5             return i;
6         }
7     }
8     return -1;
9 }
10 int main() {
11     int arr[] = {10, 20, 30, 40, 50};
12     int n = sizeof(arr) / sizeof(arr[0]);
13     int key = 30;
14     int result = linearSearch(arr, n, key);
15     if (result != -1) {
16         printf("Element found at index %d\n", result);
17     } else {
18         printf("Element not found\n");
19     }
20     return 0;
21 }
```

Test Input

```
5
10 20 30 40 50
30
```

Output

```
Element found at index 2
```

Pending manual review

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Q1

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Q2

2. Buggy Code in an array

1 mark

Q3

Identify an error in the following snippet.

1 mark

3/3 answered

Q2 2. Buggy Code in an array

1 mark

```
#include <stdio.h>

int main() {
    int arr[5] = {10,20,30,40,50};
    for(int i=0;i<=5;i++)
        printf("%d ",arr[i]);
    return 0;
}
```

Expected output:
10 20 30 40 50

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int main() {
3     int arr[5] = {10, 20, 30, 40, 50};
4     for (int i = 0; i < 5; i++) {
5         printf("%d ", arr[i]);
6     }
7     return 0;
8 }
```

Test Input

stdin input (optional)

Output

10 20 30 40 50

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Submitted

QUESTIONS

Q1

1. Find the bugs, include the missing lines in the following snippet.
1 mark

Q2

2. Buggy Code in an array
1 mark

Q3

Identify an errors in the following snippet.
1 mark

3/3 answered

Q3 Identify an errors in the following snippet

1 mark

```
int binarySearch(int arr[], int n, int key)
{
    int low = 0, high = n - 1;
    while(low < high)
    { int mid = (low + high) / 2;
      if(arr[mid] == key) return mid;
      else if(arr[mid] < key)
        low = mid;
      else high = mid;
    }
    return -1;
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int binarySearch(int arr[], int n, int key) {
3     int low = 0, high = n - 1;
4     while (low <= high) {
5         int mid = (low + high) / 2;
6         if (arr[mid] == key) {
7             return mid;
8         } else if (arr[mid] < key) {
9             low = mid + 1;
10        } else {
11            high = mid - 1;
12        }
13    }
14    return -1;
15 }
16 int main() {
17     int arr[] = {10, 20, 30, 40, 50};
18     int n = sizeof(arr) / sizeof(arr[0]);
19     int key = 30;
20     int result = binarySearch(arr, n, key);
21     if (result != -1)
22         printf("Element found at index %d\n", result);
23     else
24         printf("Element not found\n");
25     return 0;
26 }
```

Test Input

```
5
10 20 30 40 50
30
```

Output

```
Element found at index 2
```

Pending manual review